AWS CLOUD • PART V

VPC & Peering

Connecting isolated AWS networks with VPC peering

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00 VPC Recap

A Virtual Private Cloud (VPC) is your own isolated network inside AWS. It provides the network environment where resources such as EC2 instances and RDS databases can run and communicate.

The pieces that work together

A VPC is not a firewall. Instead, several networking components work together:

Component	What it does
VPC	Defines the private network and its IP address range.
Subnet	Divides the VPC into smaller networks. Each subnet belongs to one Availability Zone.
Route table	Determines where network traffic can travel.
Internet Gateway	Provides a path between the VPC and the internet.
Public IP	Gives a resource, such as an EC2 instance, an internet-routable address.
Security Group	Acts as a virtual firewall, controlling allowed inbound and outbound traffic.

For example, a public EC2 web server might live inside a subnet with a route to an Internet Gateway. The instance can have a public IP, while its Security Group allows HTTP traffic on port 80.

The important idea

No single setting provides internet access by itself. The network, routing, addressing and security rules must work together.

A VPC can contain multiple subnets

A VPC provides a larger IP address range that can be divided into smaller subnet ranges. Each subnet then provides private IP addresses for the resources placed inside it.

Network	CIDR range
VPC	10.10.0.0/16
Subnet A	10.10.1.0/24
Subnet B	10.10.2.0/24
Subnet C	10.10.3.0/24

A subnet is a range of IP addresses that resources in that subnet can use.

For example, resources inside 10.10.1.0/24 could receive private addresses such as 10.10.1.10, 10.10.1.11 and 10.10.1.12.

Subnets and route tables

Each subnet is associated with a route table. Multiple subnets can share the same route table when they need the same routing behaviour.

Private subnet

Destination	Target
10.10.0.0/16	local

This route allows traffic to stay within the VPC, but there is no direct route to an Internet Gateway. We call this a private subnet.

Public subnet

Destination	Target
10.10.0.0/16	local
0.0.0.0/0	Internet Gateway

The second route provides a direct path to an Internet Gateway. A subnet with this routing configuration is considered a public subnet.

How VPC networking fits together

A VPC is the larger private network. Subnets divide its IP range into smaller networks, and resources such as EC2 instances receive private IP addresses from the subnet they belong to.

Each subnet uses a route table to determine where traffic can be sent. For example:

Destination	Target	Meaning
10.10.0.0/16	local	Traffic within the VPC stays within the VPC.
0.0.0.0/0	Internet Gateway	Other IPv4 traffic has a path toward the internet.

0.0.0.0/0 represents all IPv4 destinations.
When its target is an Internet Gateway, the subnet has a route toward the internet. However, that route alone does not give an EC2 instance direct IPv4 internet connectivity.

The EC2 instance also needs a public IPv4 address. Its private IP is used inside the VPC, while the public IP provides the internet-routable address needed for direct external communication.

Route table: determines where the subnet's traffic can be routed.

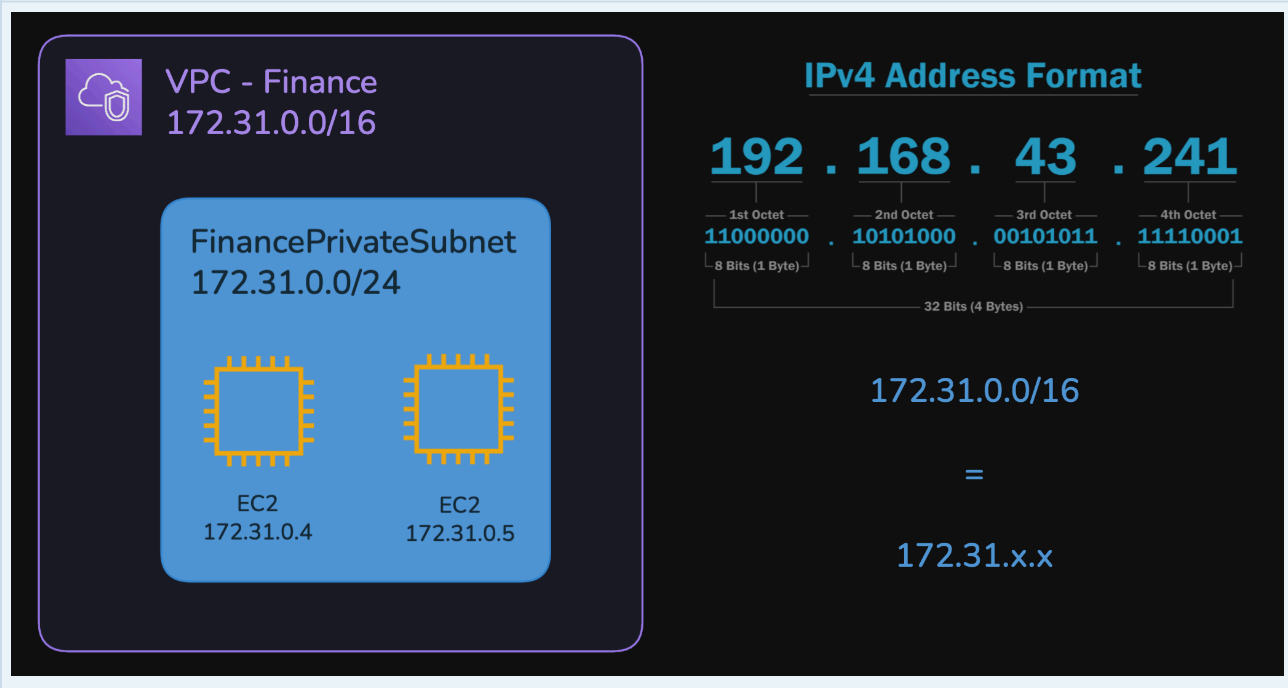
EC2 addressing: determines whether the individual instance has the public address needed to communicate directly with the internet.

This means an EC2 instance can have only a private IP address while still being located inside a public subnet. The subnet is public because of its routing configuration, not because every resource inside it has a public IP.

In short: the VPC provides the network, subnets divide that network, route tables provide routing instructions, and EC2 instances live inside the subnets and use those routes when communicating.

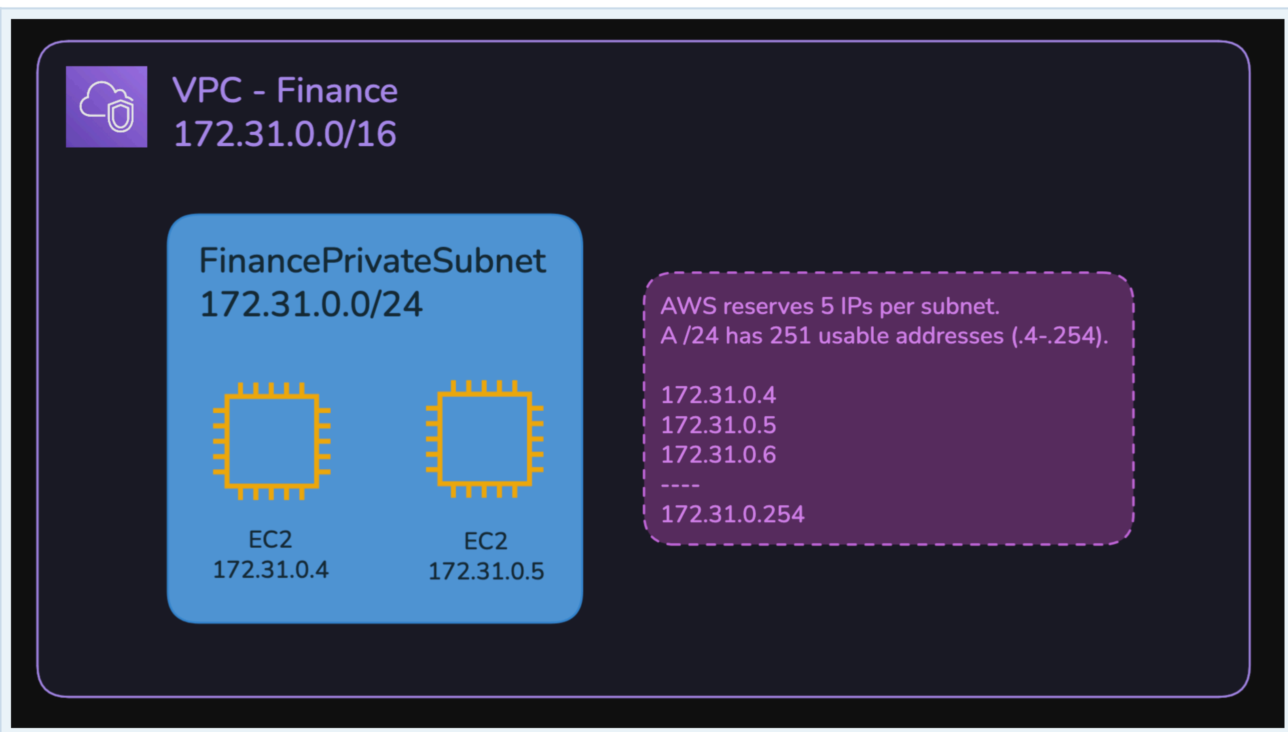
VPC & CIDR Size

A VPC is given a range of private IP addresses using CIDR notation. A /16 fixes the first two octets, leaving the last two available for addressing. For example, 172.31.0.0/16 covers 172.31.0.0 to 172.31.255.255, giving 65,536 total addresses.



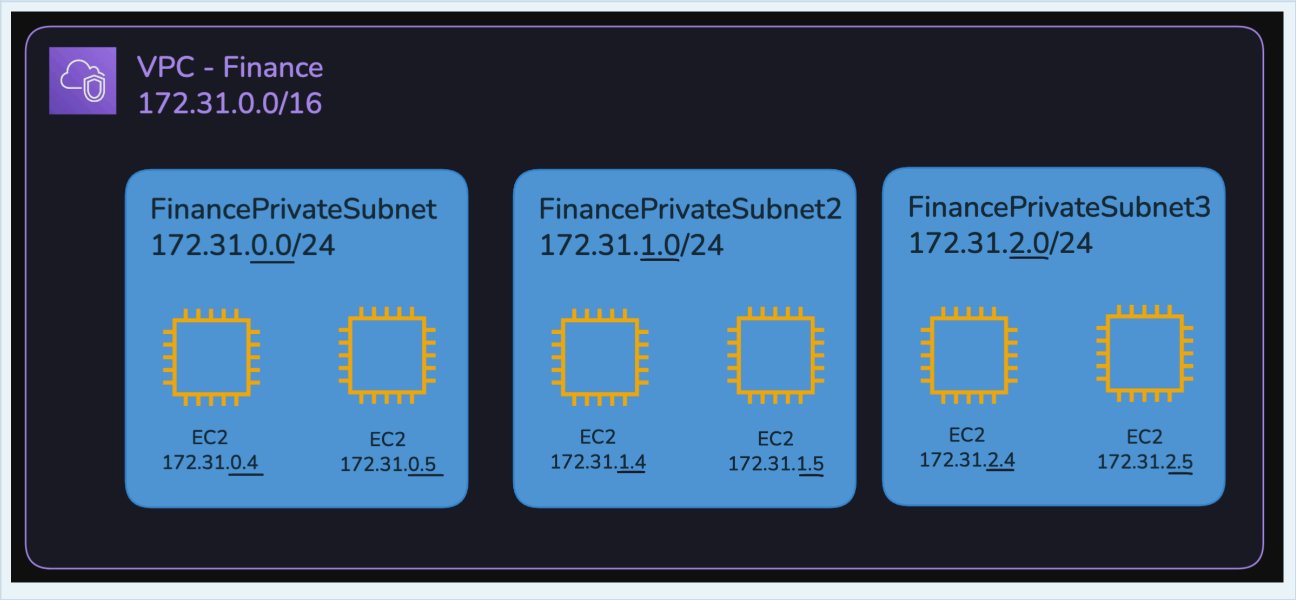
Subnets & Available Addresses

A subnet divides the larger VPC address range into smaller networks. A /24 subnet contains 256 addresses. AWS reserves five addresses in each subnet, leaving 251 usable addresses, such as 172.31.0.4 through 172.31.0.254.



Multiple Subnets

A VPC can contain many subnets, each using a different portion of the VPC address range. For example, 172.31.0.0/24, 172.31.1.0/24 and 172.31.2.0/24 are separate networks, but all belong within the larger 172.31.0.0/16 VPC.



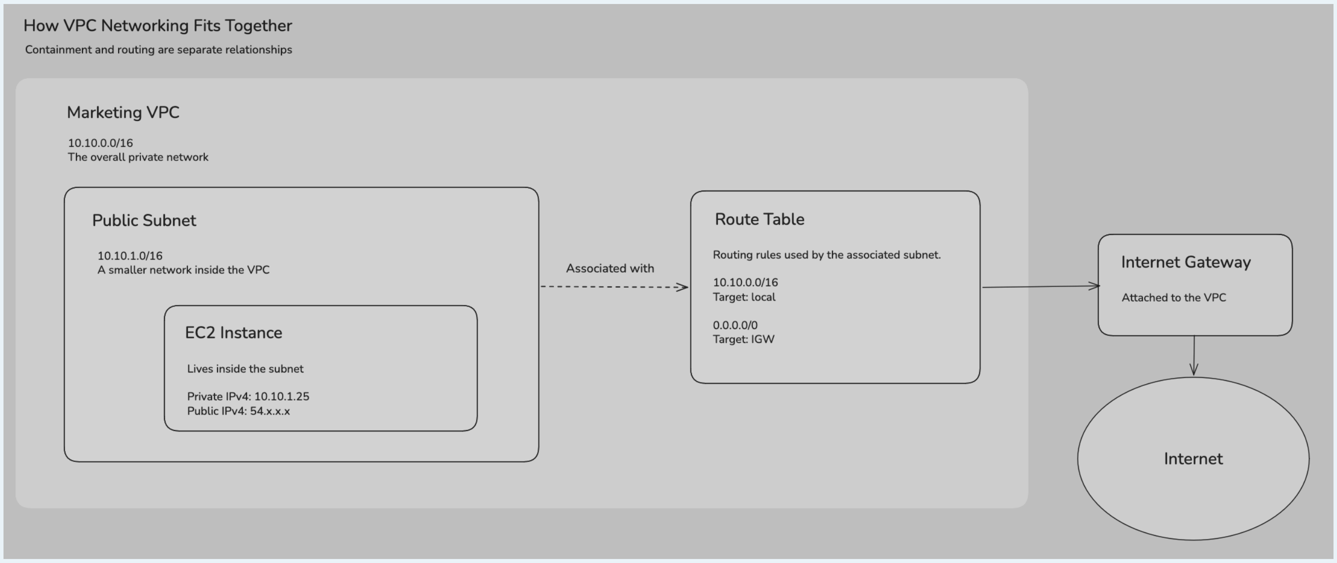
Think about it:
If the VPC already has the CIDR block 172.31.0.0/16, why do the subnets need their own CIDR blocks?

The VPC CIDR defines the **overall address space available to the VPC**.
The subnets divide that larger space into smaller networks.
Resources such as EC2 instances are then placed inside a subnet and receive a private IP address from **that subnet's range**.

The VPC defines the address space you have available.
The subnets decide how that space is divided and used.

Subnets & Availability Zones

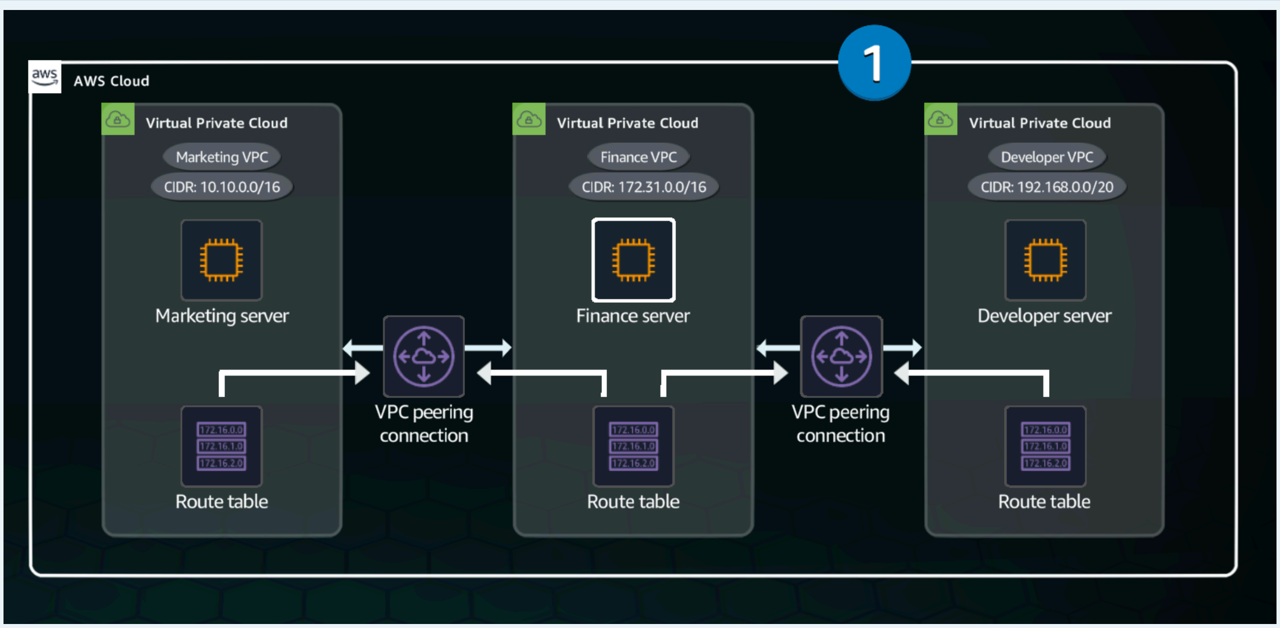
Multiple subnets can exist within the same Availability Zone. However, each individual subnet belongs to only one AZ and cannot span multiple AZs. Creating subnets in different AZs allows resources to be distributed for higher availability.



01 Connecting VPCs

Overview

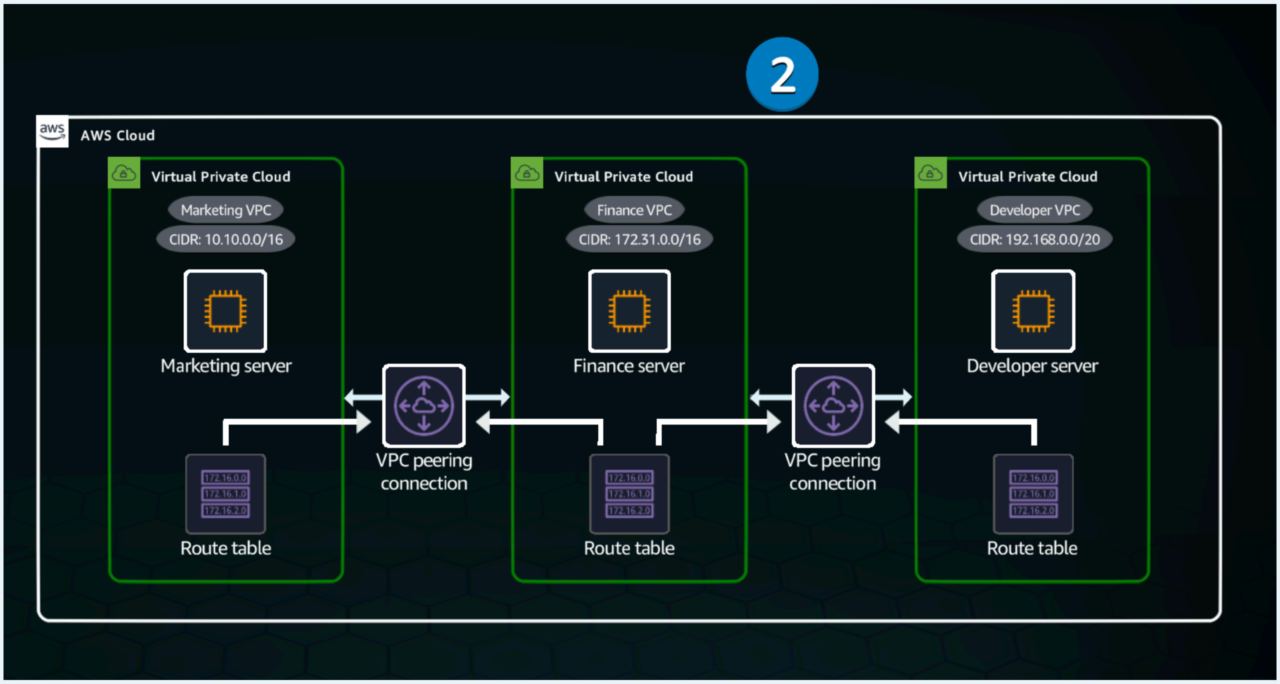
This solution involves three distinct Virtual Private Clouds (VPCs) for the Marketing, Finance and Development departments. The Finance VPC hosts resources required by both the Marketing and Development VPCs.



VPCs are isolated by default

Resources in separate VPCs cannot communicate simply because they are in AWS. Access between VPCs requires an explicit network connection.

VPC peering creates a direct private connection between two VPCs through the AWS network. It does not require an Internet Gateway, VPN connection or physical networking hardware.



VPC peering and CIDR blocks

Important: Peered VPCs must use non-overlapping CIDR blocks. Two VPCs that both use 192.168.0.0/24 cannot be peered.

Each VPC has a CIDR block that defines its private IP range. That range can then be divided into smaller ranges for subnets.

VPC	CIDR block
Marketing	10.10.0.0/16
Finance	10.20.0.0/16
Development	10.30.0.0/16

Each VPC can create subnets from its own range. For example, Marketing could use 10.10.1.0/24 and 10.10.2.0/24, while Finance could use 10.20.1.0/24 and 10.20.2.0/24.

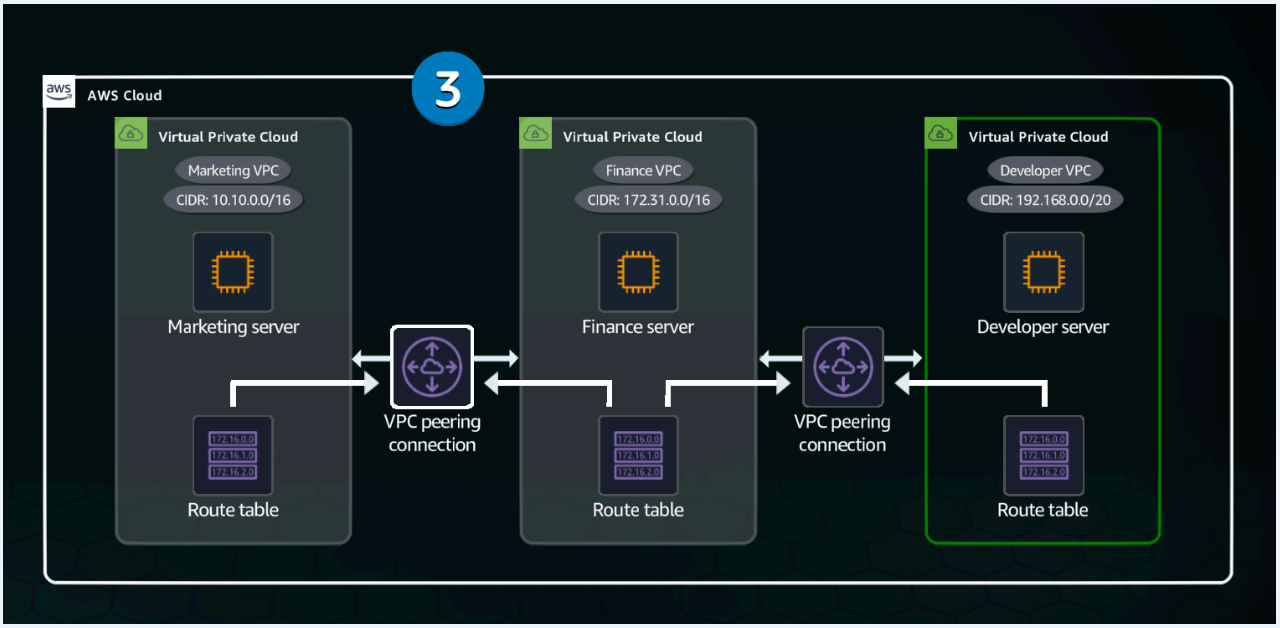
Reusing private IP ranges in completely isolated networks is normally fine. The problem appears when those networks need to communicate: routing must be able to tell which VPC owns the destination address.

For example, Marketing can route 10.10.0.0/16 locally and send 10.20.0.0/16 through the Finance peering connection. Because the ranges are different, AWS knows exactly where the traffic belongs.

Key point: Private IP ranges can be reused across isolated networks, but VPCs connected through peering must use non-overlapping CIDR blocks so traffic can be routed to the correct network.

Marketing connects to Finance

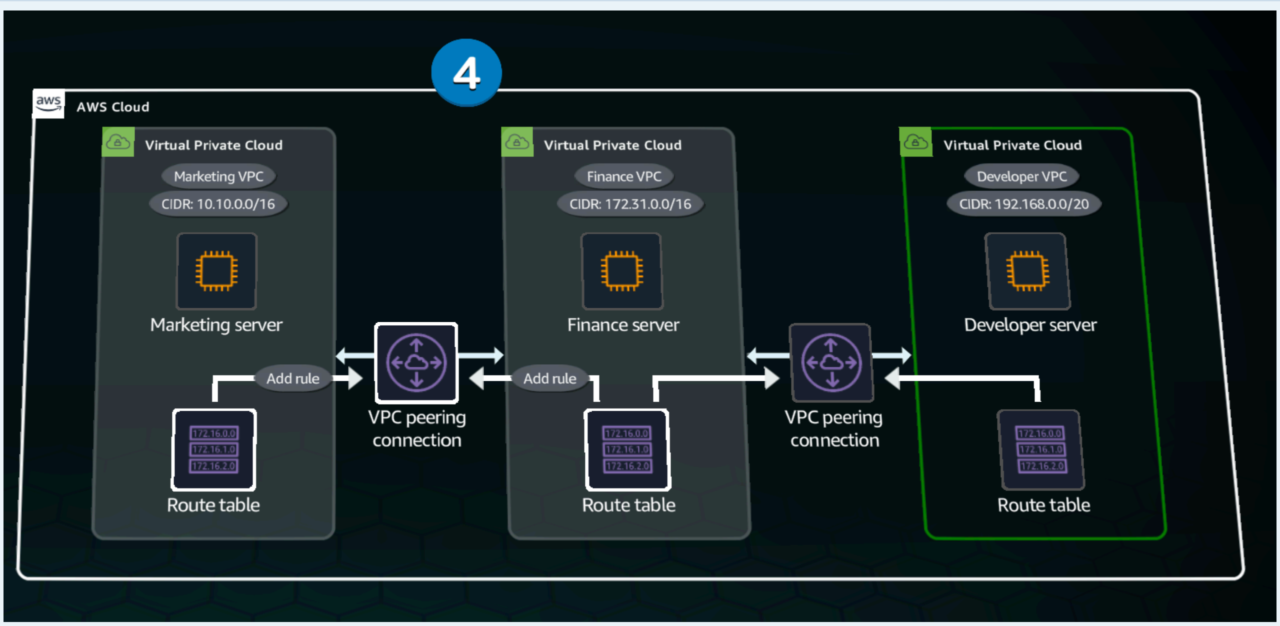
A VPC peering connection is established between the Marketing and Finance VPCs, enabling private IPv4 or IPv6 traffic to be routed between them.



Routing must work in both directions

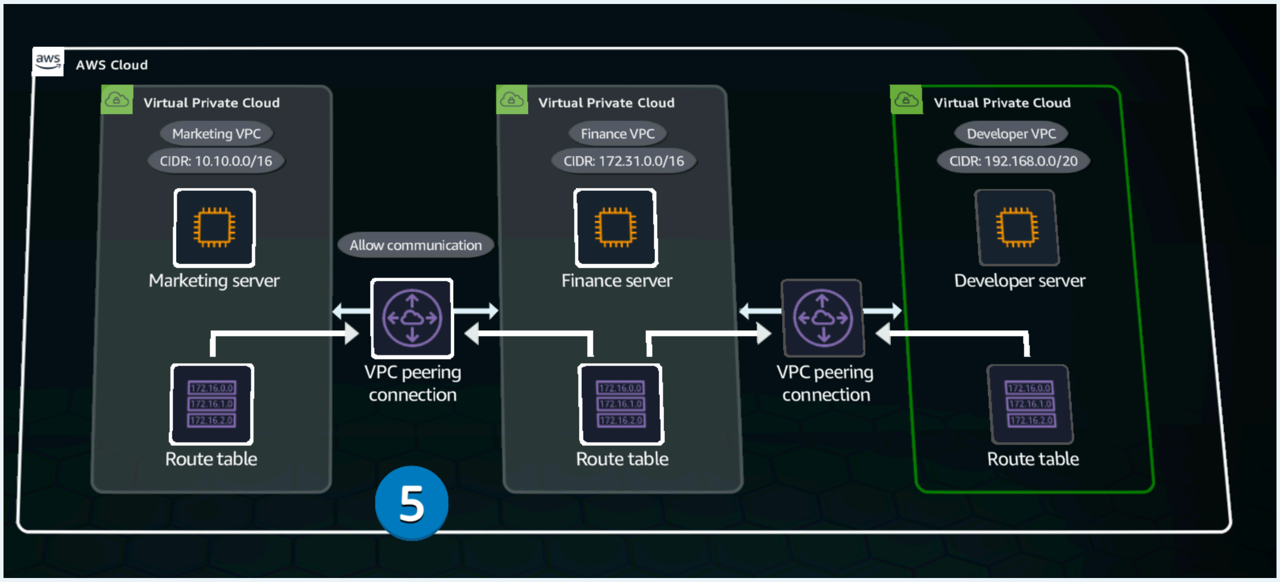
Creating the peering connection is not enough. Route tables in both VPCs need routes that point traffic for the other VPC to the peering connection.

The Marketing route table sends Finance-bound traffic through the peering connection, while the Finance route table provides the return path back to Marketing.



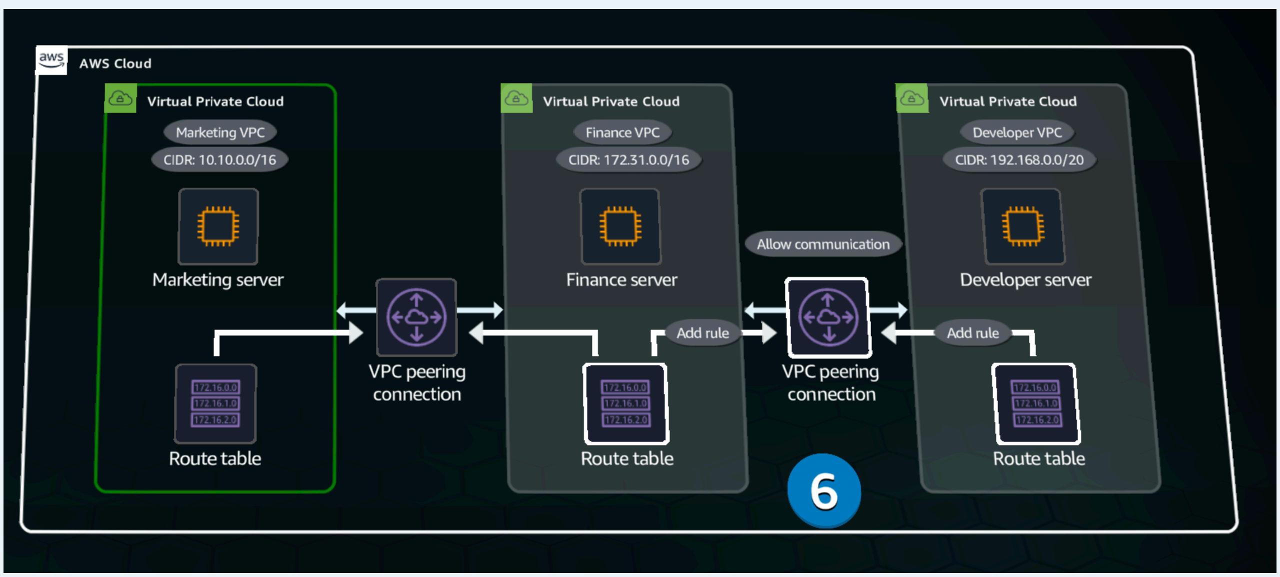
The VPCs can now communicate

Once the peering connection is active and the required routes and security rules are configured, resources in Marketing and Finance can communicate using their private addresses.



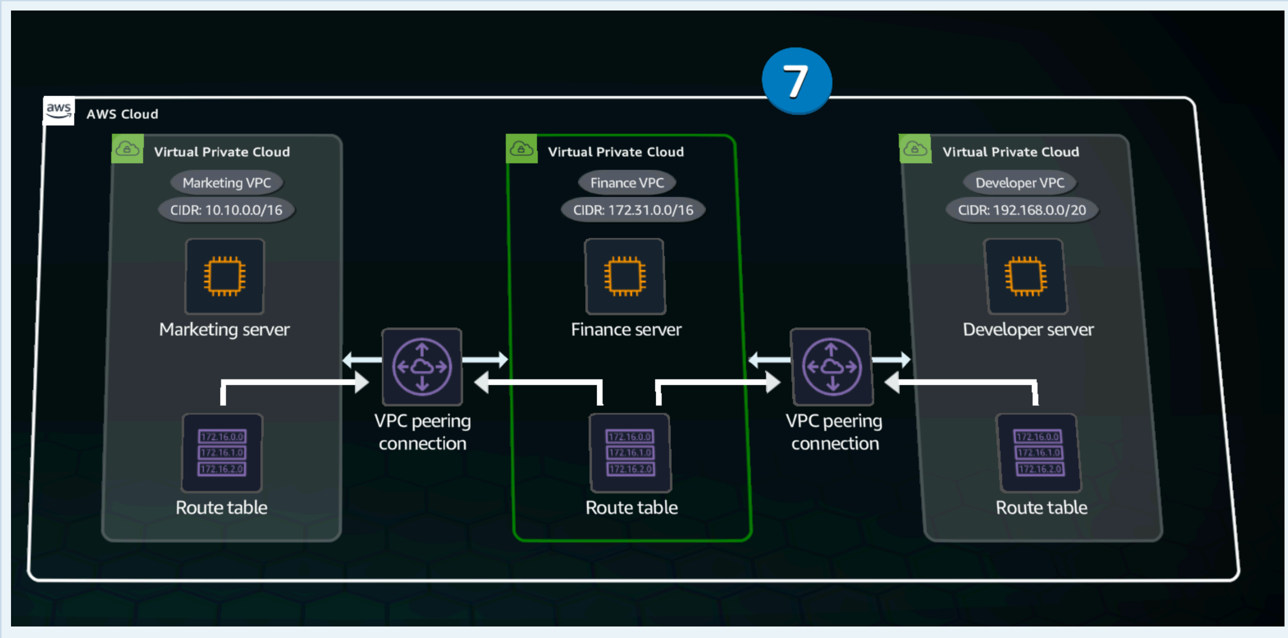
Development also needs Finance

Development requires its own separate peering connection to Finance, with corresponding route table updates in both VPCs.



Why Marketing and Development remain isolated

No peering connection exists between Marketing and Development because they do not require direct communication. VPC peering is not transitive, so traffic cannot travel from Marketing through Finance and then into Development.



Why can’t they use the same peering connection?

A VPC peering connection connects exactly two VPCs. It has two endpoints: a requester VPC and an accepter VPC. Marketing and Finance therefore need one peering connection, while Development and Finance need another.

Those two connections still do not let Marketing and Development communicate through Finance. Finance cannot act as a middleman because VPC peering does not support transitive routing.

When peering becomes difficult to manage

If many VPCs need to connect through a central networking hub, maintaining many individual peering connections can become cumbersome. AWS Transit Gateway is designed for this type of architecture.

02 Connecting VPCs

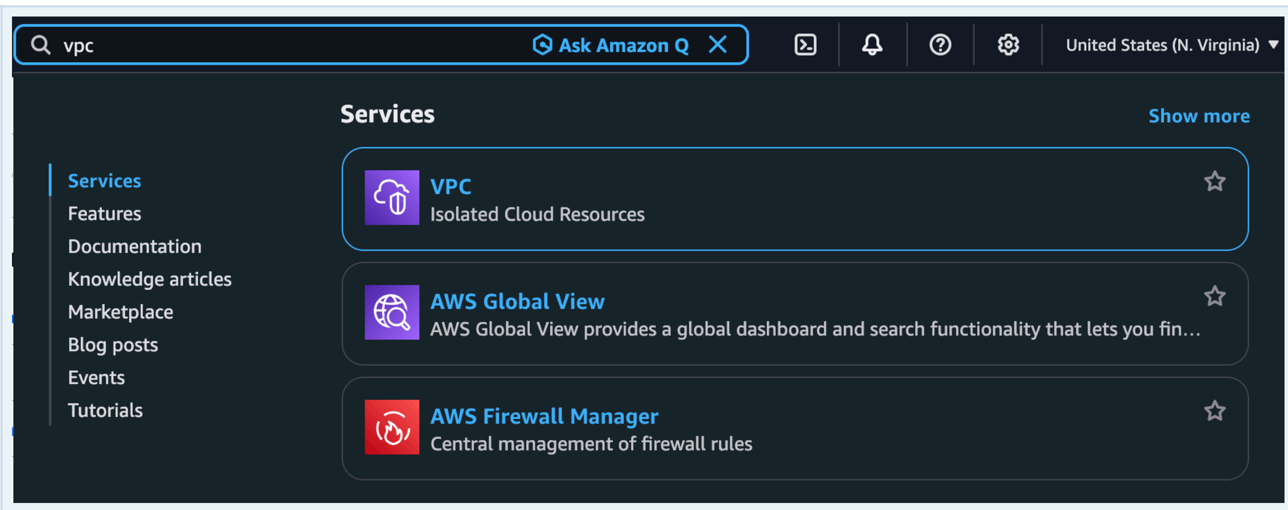
Why can’t both VPCs use the same peering connection?

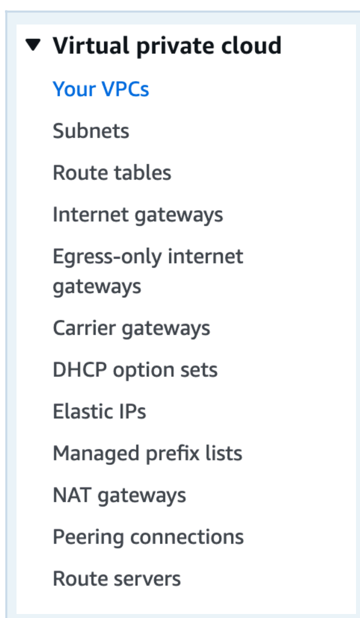
A VPC peering connection connects exactly two VPCs: a requester and an accepter. Marketing and Finance therefore need one connection, while Development and Finance need another. Finance cannot act as a middleman between them because VPC peering is not transitive. If many VPCs need a central networking hub, AWS Transit Gateway is designed for that type of architecture.

Practice Lab

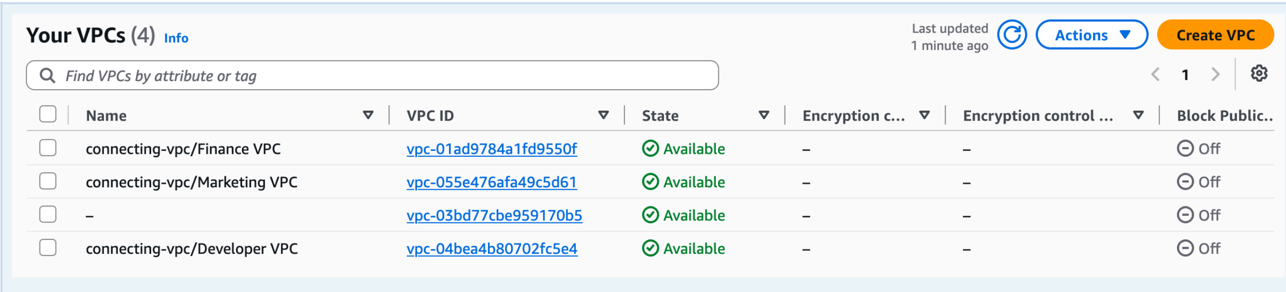
Goals: set up a VPC peering connection and make sure traffic is properly routed between the peered VPCs.

Amazon Virtual Private Cloud (VPC) lets us launch AWS resources inside a logically isolated virtual network that we define and control.



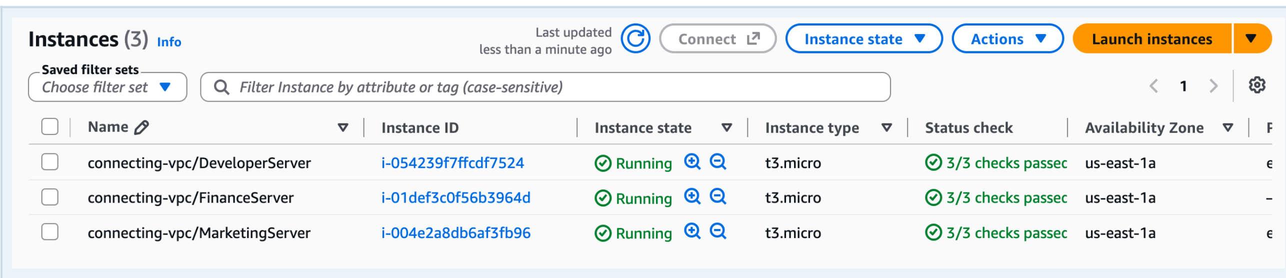


Here we can see the Finance, Marketing and Development VPC structure described earlier. By default, these VPCs are isolated and cannot communicate with one another.

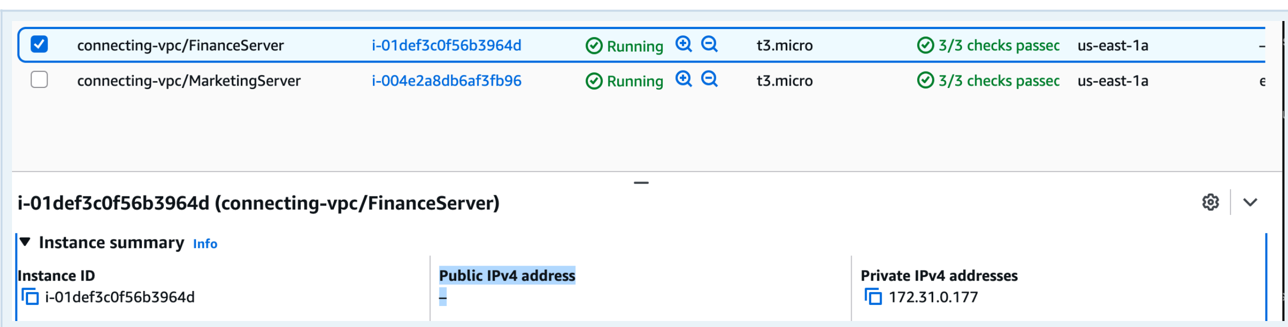


Verify the EC2 network placement

VPC peering lets resources in two separate VPCs communicate using their private IP addresses. Every EC2 instance runs inside a VPC and a subnet. In the EC2 console, we can verify each instance's VPC, subnet and private IP address and confirm that Marketing, Finance and Development belong to their respective VPCs.

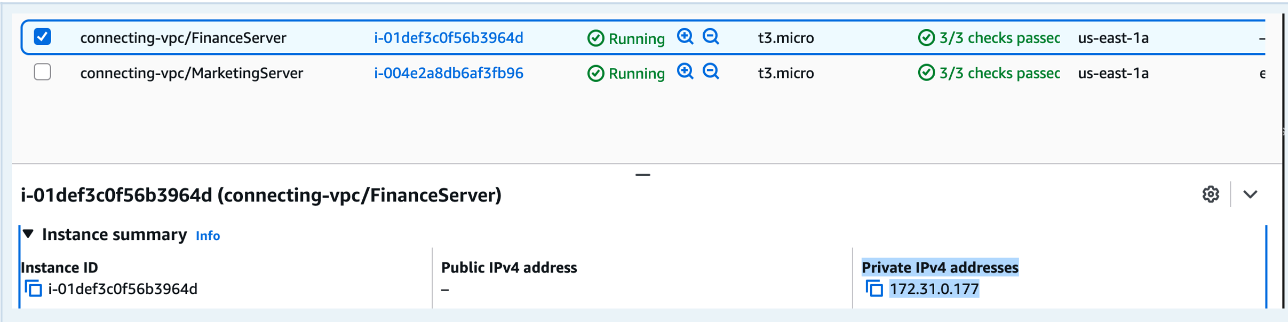


A subnet exists entirely within one Availability Zone. In this lab, the Finance instance has no public IP address or public DNS name, confirming that we must reach it through private network connectivity.



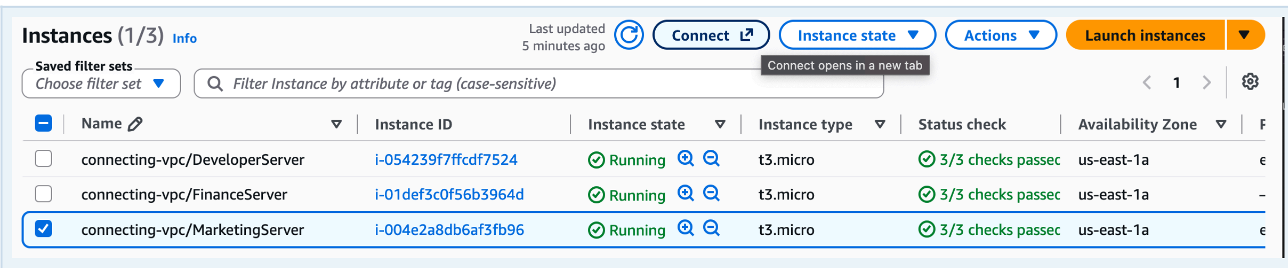
Record the Finance private IP

On the Finance EC2 instance, copy the Private IPv4 address and save it. We will use this address when testing connectivity from Marketing.

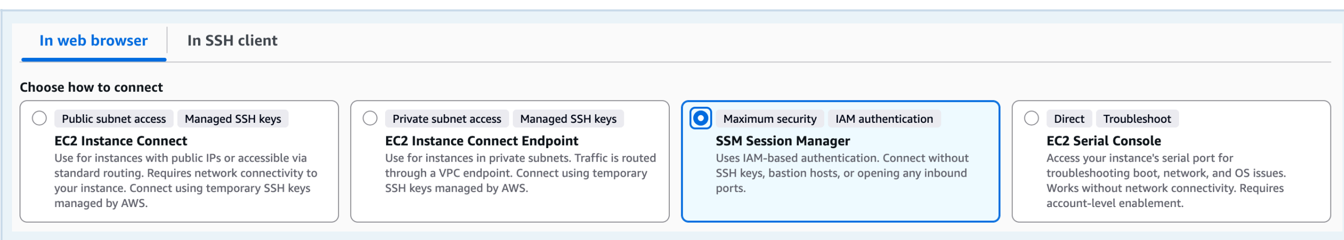


Testing connectivity: Marketing to Finance

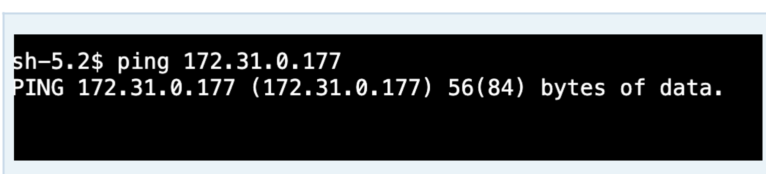
Before configuring peering, we first prove that the two isolated VPCs cannot yet communicate. Select MarketingServer and choose Connect.



Session Manager, a capability of AWS Systems Manager, provides browser-based terminal access to EC2 instances. In this lab it lets us access MarketingServer without a public IP address, SSH keys or an open SSH port. Commands entered here run on MarketingServer itself.

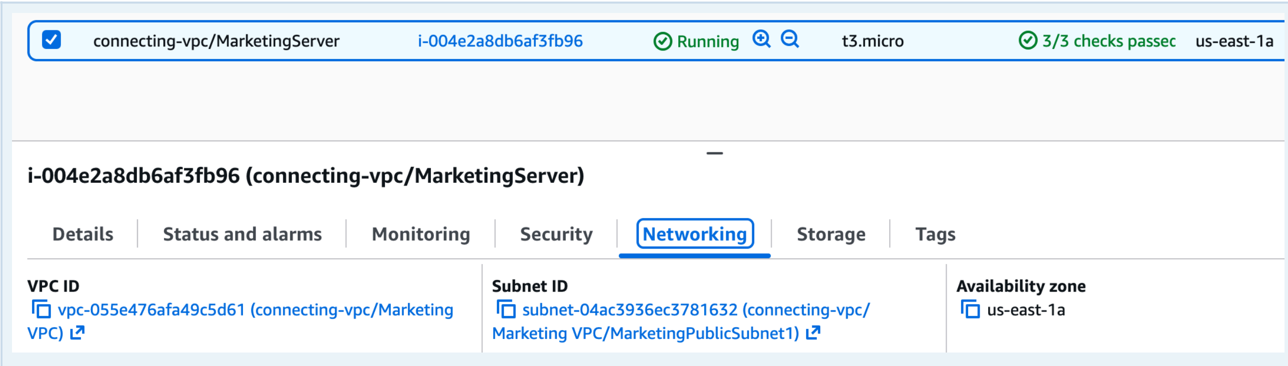


Ping the Finance private IP copied earlier. The command receives no replies because there is no route from the Marketing VPC to the Finance VPC yet. Press Ctrl+C to stop the ping.

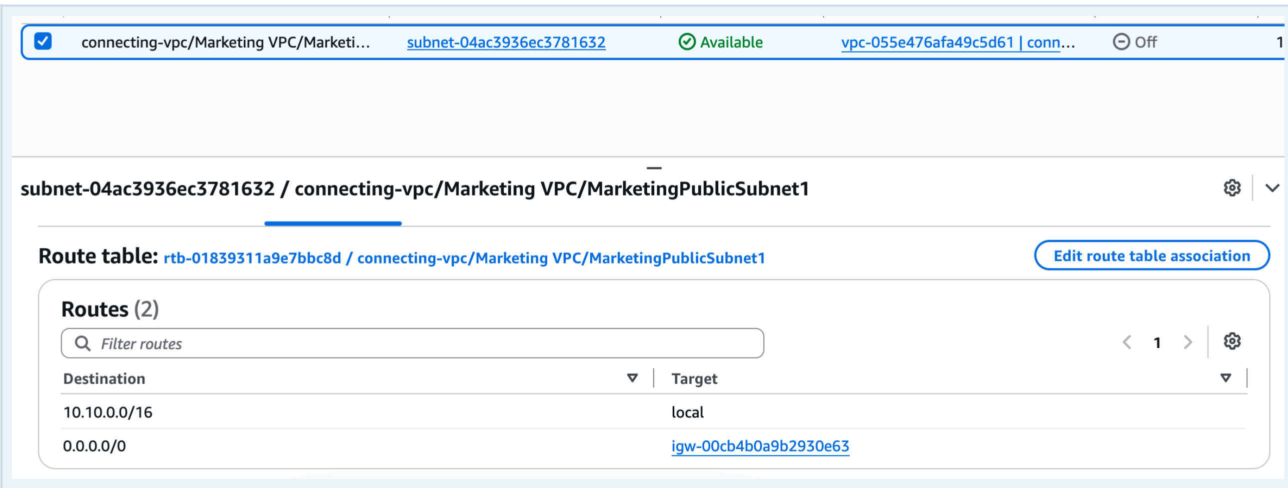


Find the Marketing route table

EC2 instances reside inside subnets. From MarketingServer, open its Subnet ID to reach the Marketing subnet.



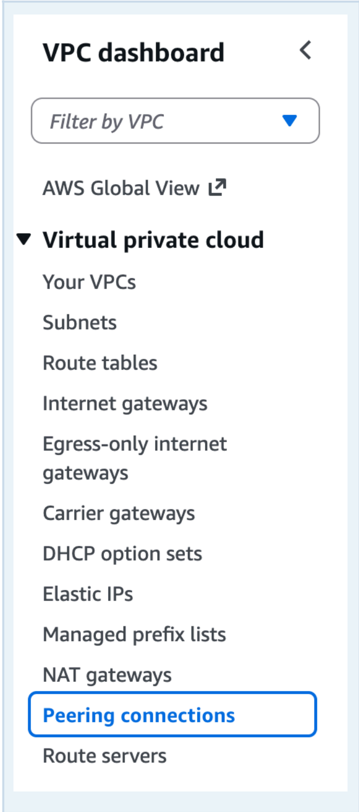
Each subnet is associated with a route table. The route table determines where traffic for different destination networks should be sent.



Private VPC networks use private ranges such as 10.x.x.x, 172.16–31.x.x or 192.168.x.x. Routes determine whether traffic stays local, travels to another VPC, or uses a gateway where one is configured.

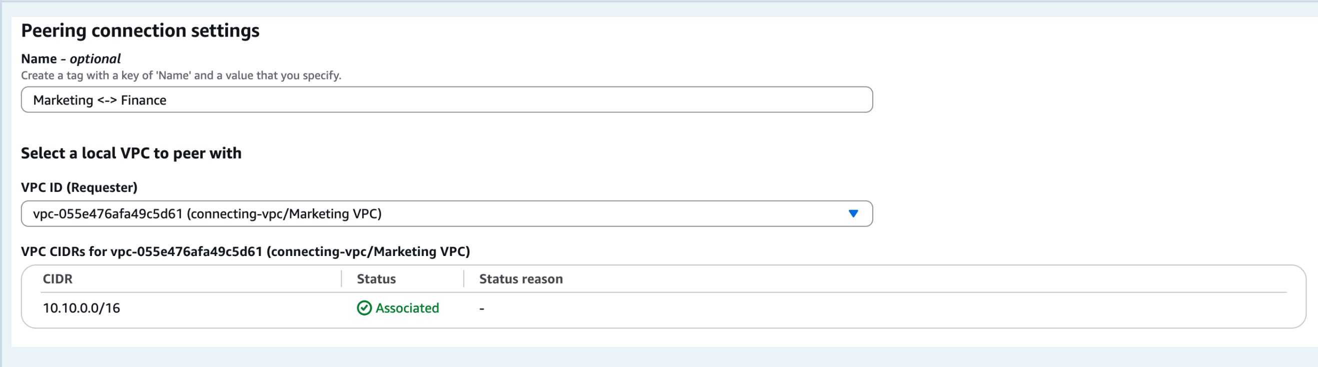
Create the VPC peering connection

The next step is to create a peering connection between Marketing and Finance.

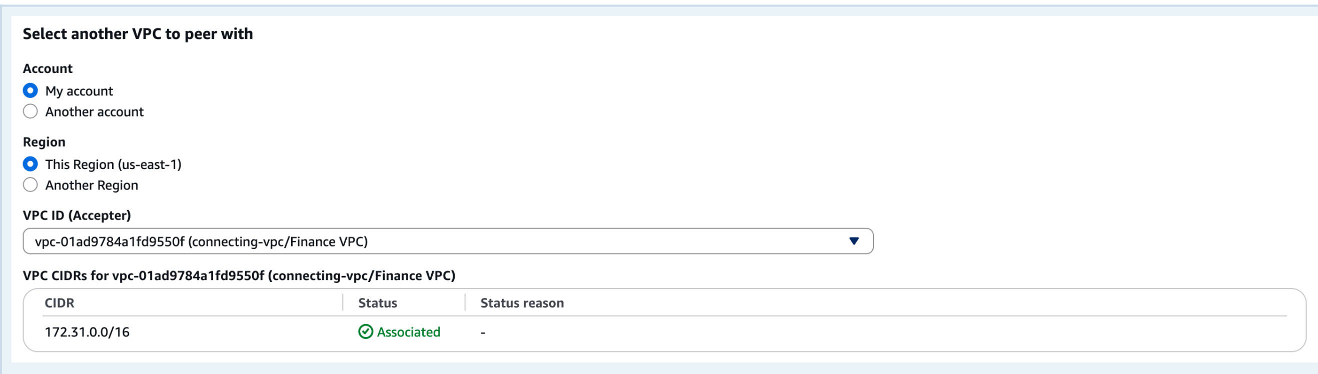


Choose Create peering connection. We want a direct private connection between the Marketing and Finance VPCs.

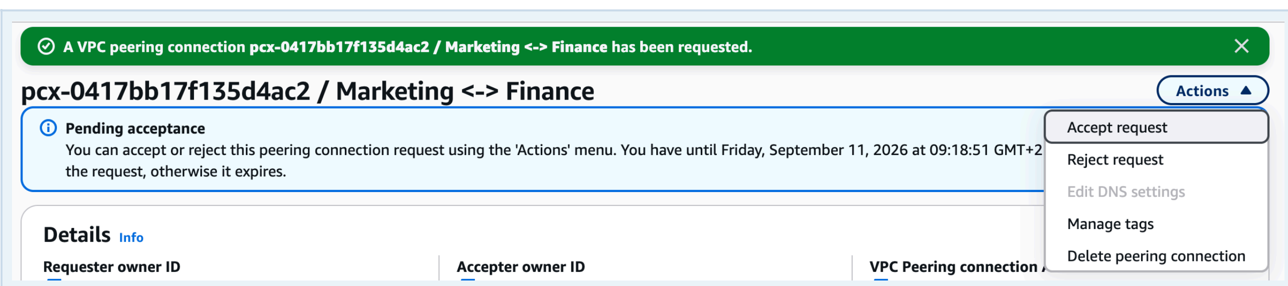
The requester VPC initiates the peering request and the accepter VPC receives it. Before creating the connection, identify both VPC IDs and their CIDR blocks, and confirm that the CIDR ranges do not overlap.



Because both VPCs are in the same AWS account, we can select the Finance VPC as the accepter and manage both sides of the request ourselves.



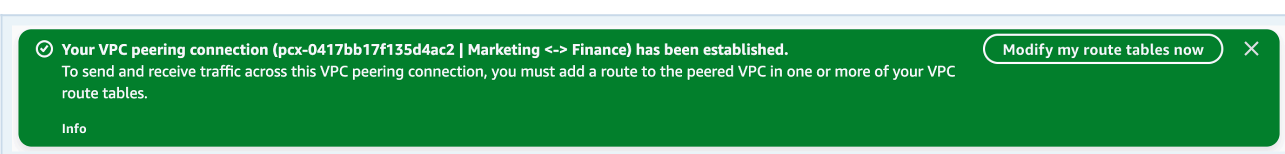
Within one account: provide both VPC IDs, create the peering request, then accept it. Finish this step by choosing Create peering connection.



Across different AWS accounts, the owner of the accepter VPC must accept the peering request.

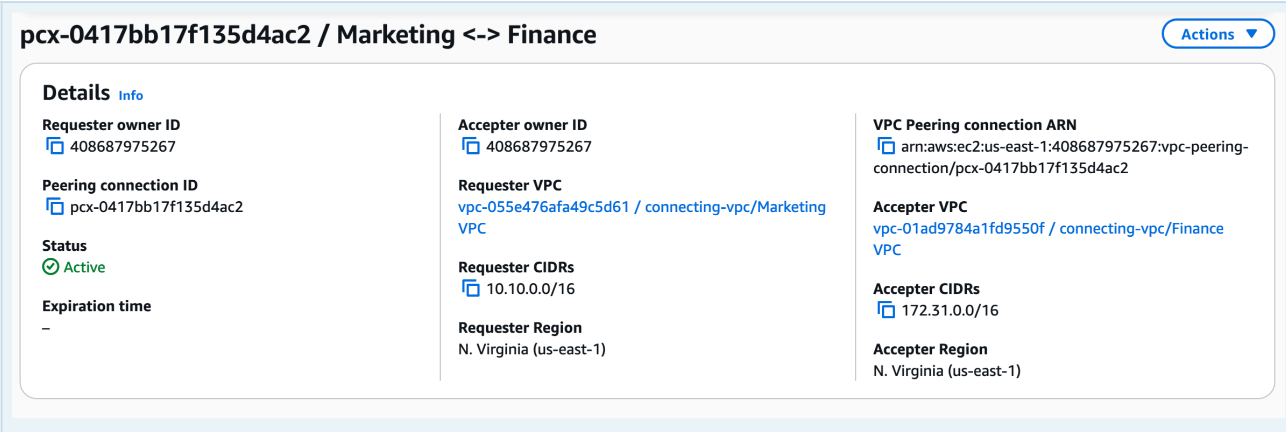
Peering accepted - routing is still required

Accepting the request creates the peering connection, but it does not automatically add the routes required for traffic to use it.

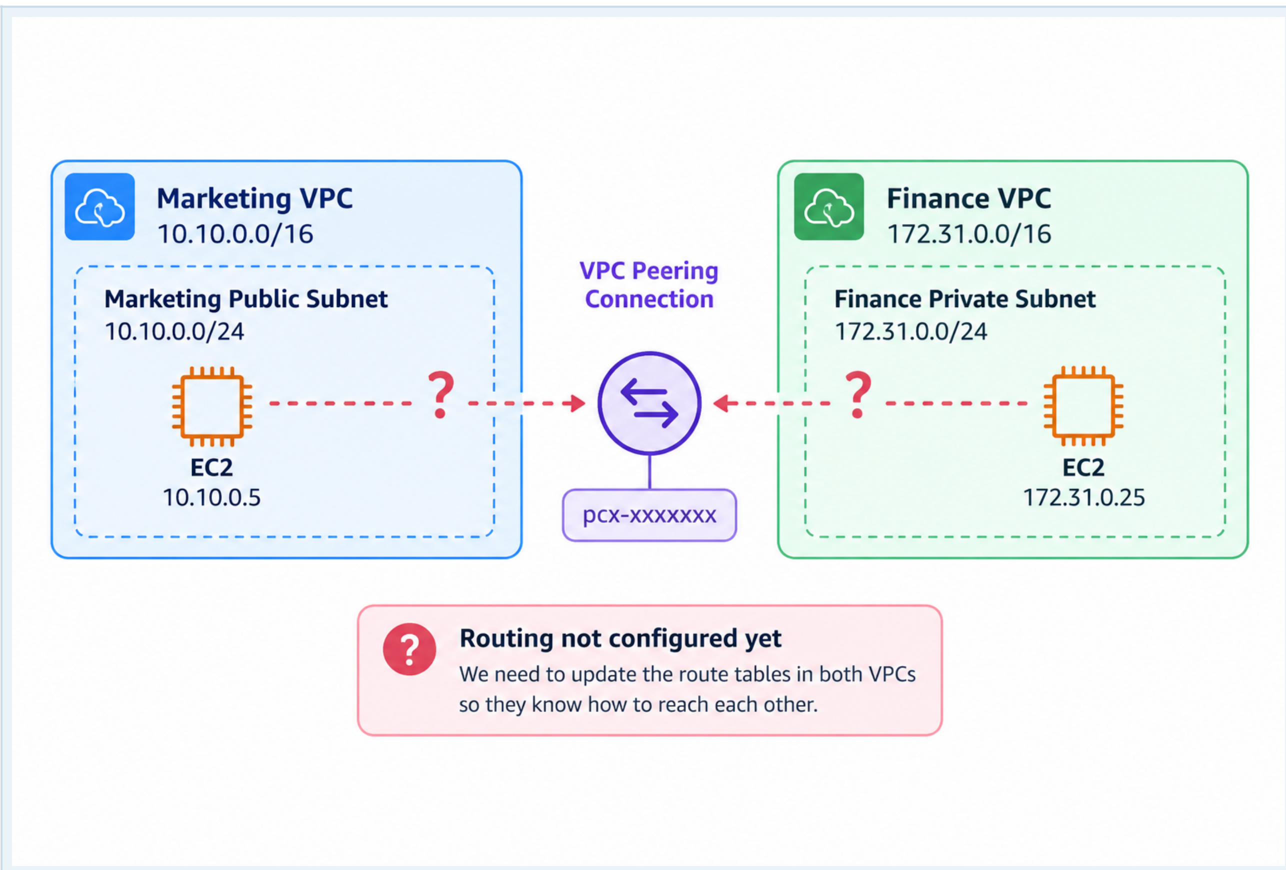


pcx is the identifier prefix for a VPC peering connection. Take note of the peering connection ID because we will use it as a route target.

Review the connection and verify that its status is Active before continuing.



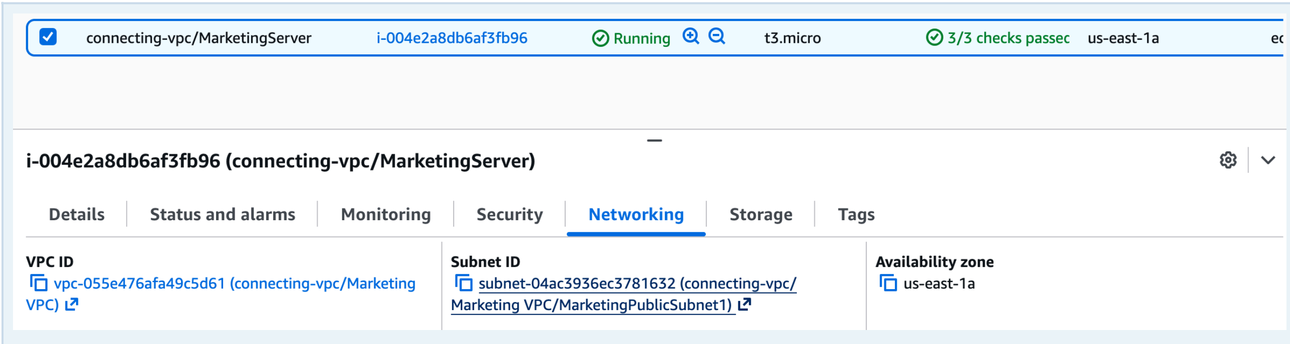
Each VPC still needs a route telling it to use the peering connection when the destination belongs to the other VPC's CIDR block.

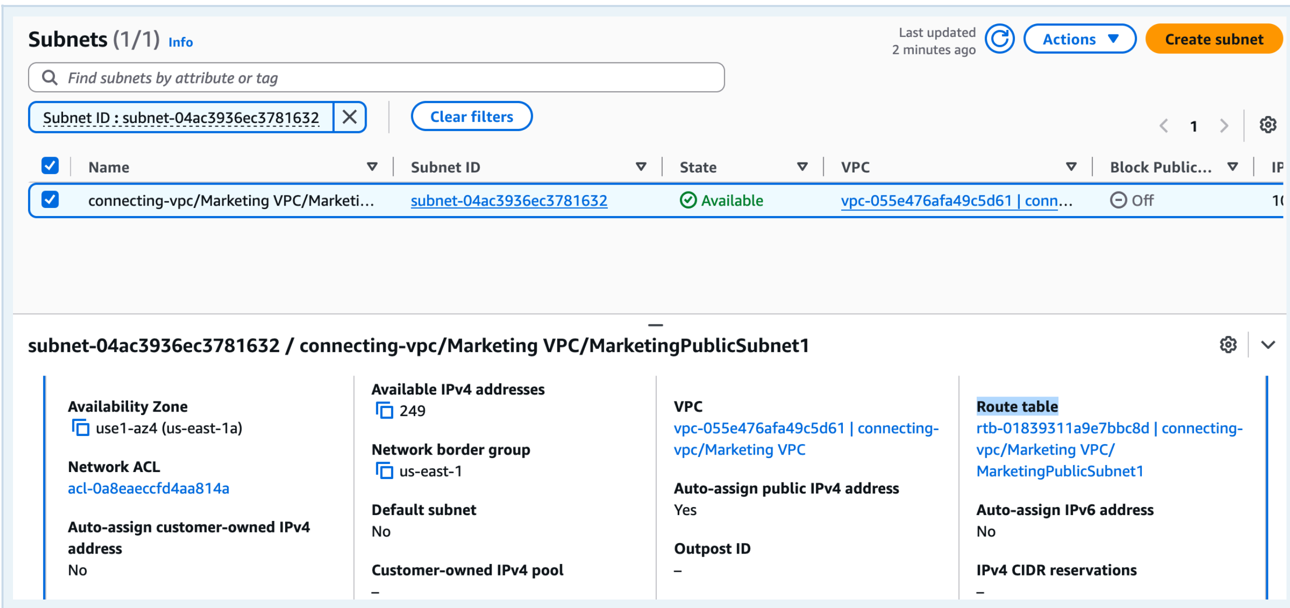


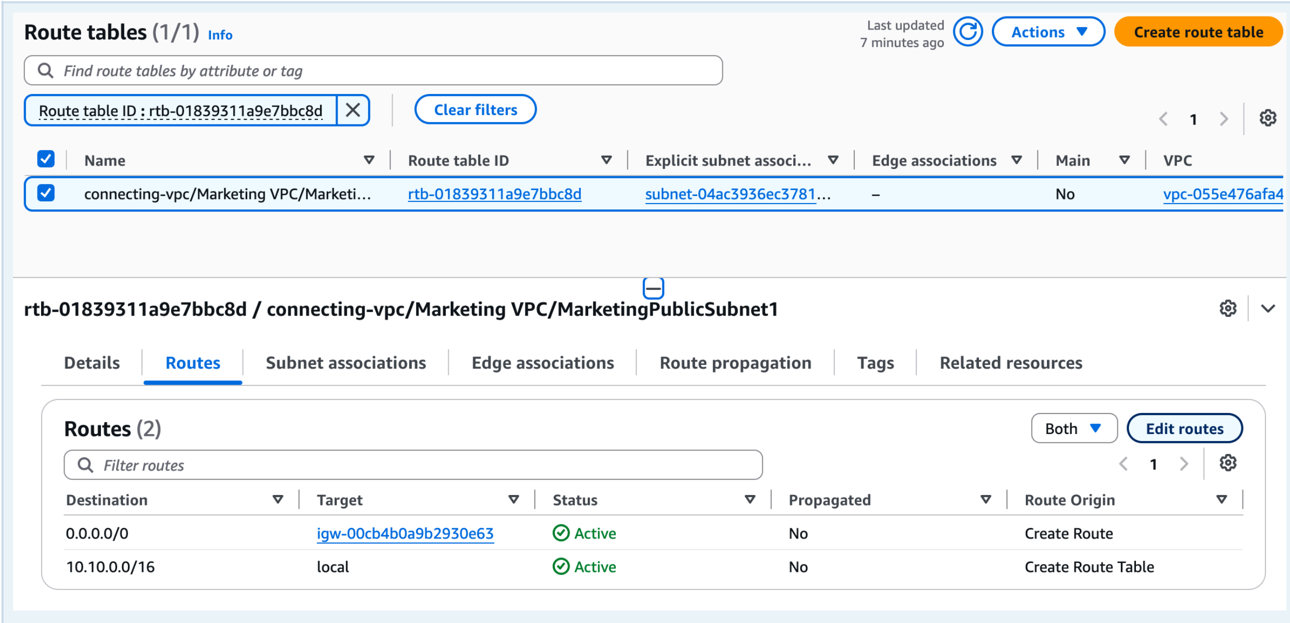
Think of the peering connection as a new road between two cities. The road exists, but the route tables provide the directions that tell traffic when to use it.

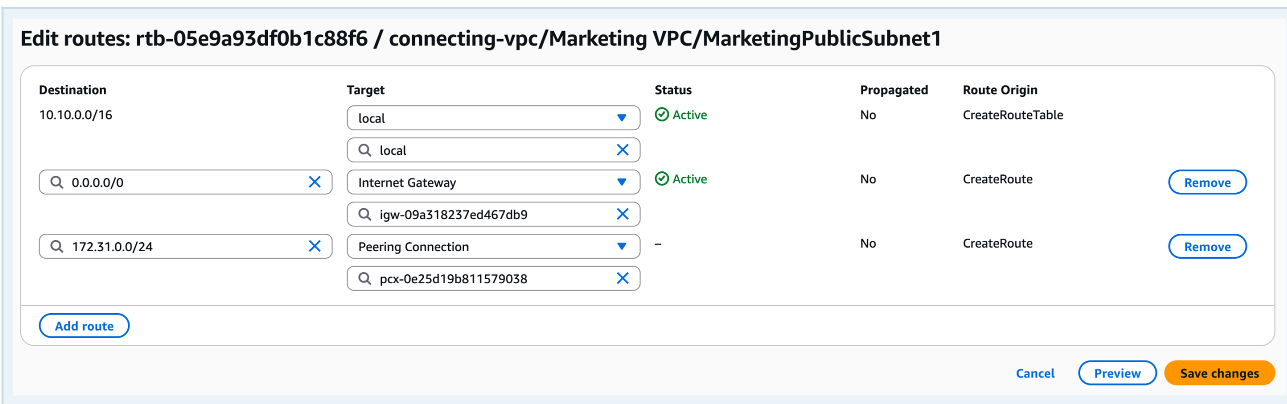
Update both route tables

Return to the Marketing subnet and open its route table. Add a route for the Finance VPC CIDR block with the VPC peering connection as the target.





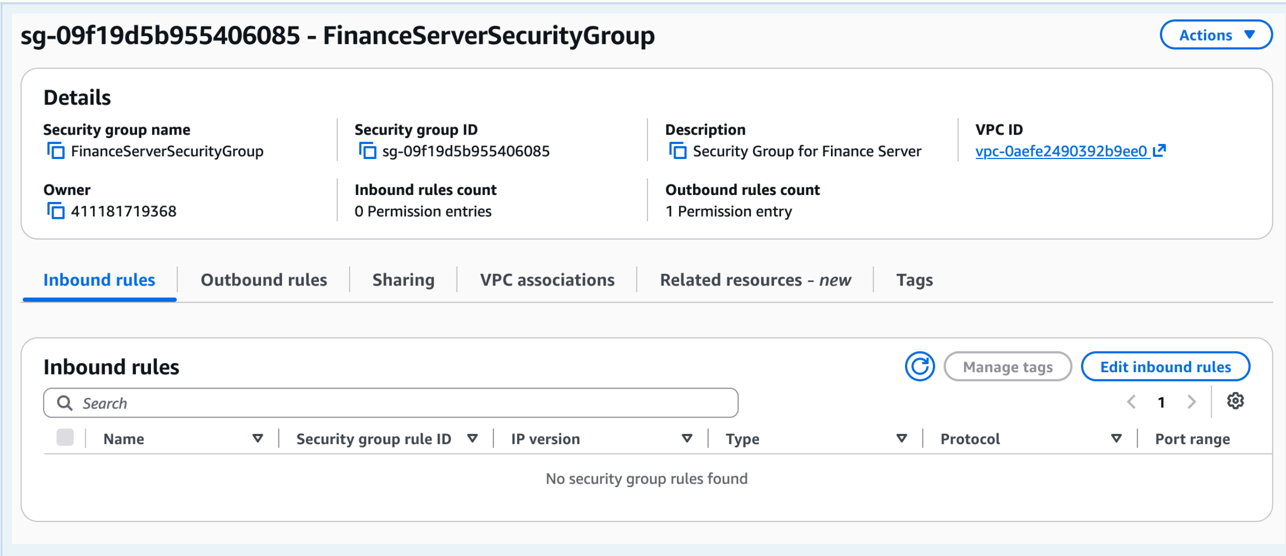


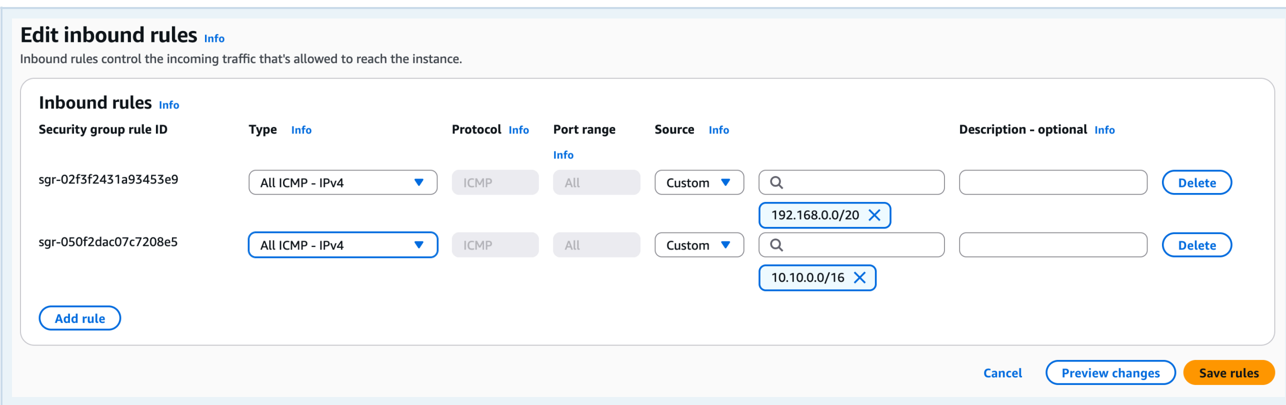


Repeat the process in Finance so traffic has a route back to Marketing. Peering requires routing in both directions.

Allow the traffic in the Security Group

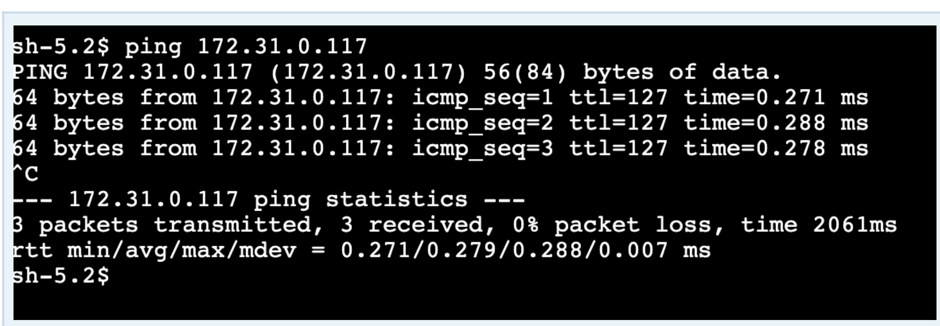
Routing provides the network path, but the relevant Security Group must also allow the traffic being tested. Update the Finance security rules as required by the lab.





Test again

Return to MarketingServer through Session Manager and ping the Finance private IP again. With the peering connection, routes and security rules in place, the FinanceServer now responds.



Success!
The two VPCs can now communicate privately through the VPC peering connection.